

9. (a) A student is given five beakers labelled **A**, **B**, **C**, **D** and **E**. Each contains a different solution. The solutions are barium bromide, barium nitrate, calcium chloride, magnesium chloride and magnesium nitrate but it is not known which beaker contains which solution.

She is also given solutions of three reagents: silver nitrate, sodium hydroxide and sulfuric acid.

Devise a plan that uses the reagents to unambiguously determine which solution is in which beaker.

You should include the observations that enable you to positively identify **all** five solutions.
[6 QER]

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- (b) Give a test which would confirm the presence of calcium ions in aqueous calcium chloride. State the result of the test. [2]

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- (c) A solution is known to contain a mixture of sodium carbonate and sodium nitrate only. To determine the carbonate ion concentration in the mixture, aqueous barium chloride was added in excess. The precipitate produced was filtered off and dried by strong heating.

- (i) Write the **ionic** equation, including state symbols, for this precipitation reaction. [1]

- (ii) State why the aqueous barium chloride was added in excess. [1]

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- (d) All Group 2 carbonates are insoluble and can be precipitated, dried and weighed in experiments similar to that in part (c). However, the strong heating needed to drive off all the water can cause a problem in accurately determining the mass of the carbonate precipitated. Suggest what this problem might be. [2]

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(e) Explain the following observations by reference to the bonding present in each of the substances.

(i) Ionic substances such as calcium chloride can conduct electricity under certain circumstances. [2]

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(ii) Iodine is a solid which vaporises on gentle warming. [2]

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(iii) A metal such as magnesium is malleable. [2]

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END OF PAPER

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